

Quantum Hardware Stack: Control Challenges for Superconducting Quantum Technology

Anastasiia Butko

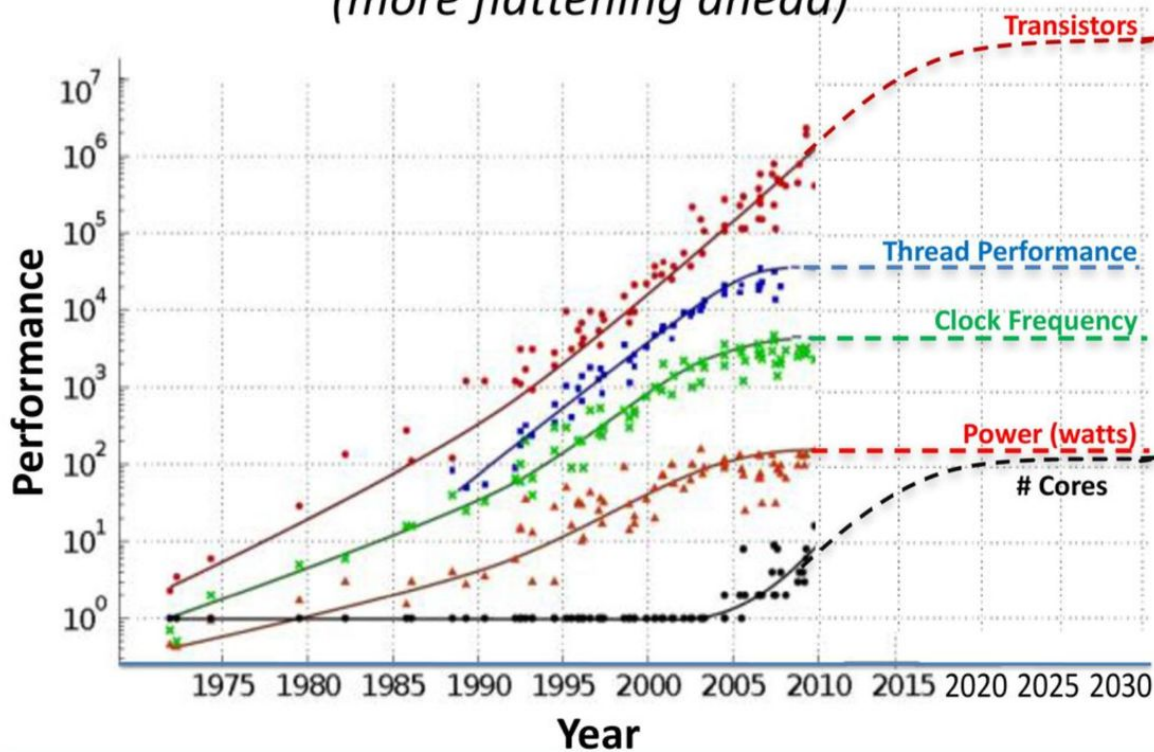
Computer Architecture Group
Lawrence Berkeley National Lab



July 13, 2026
openQSE Meeting

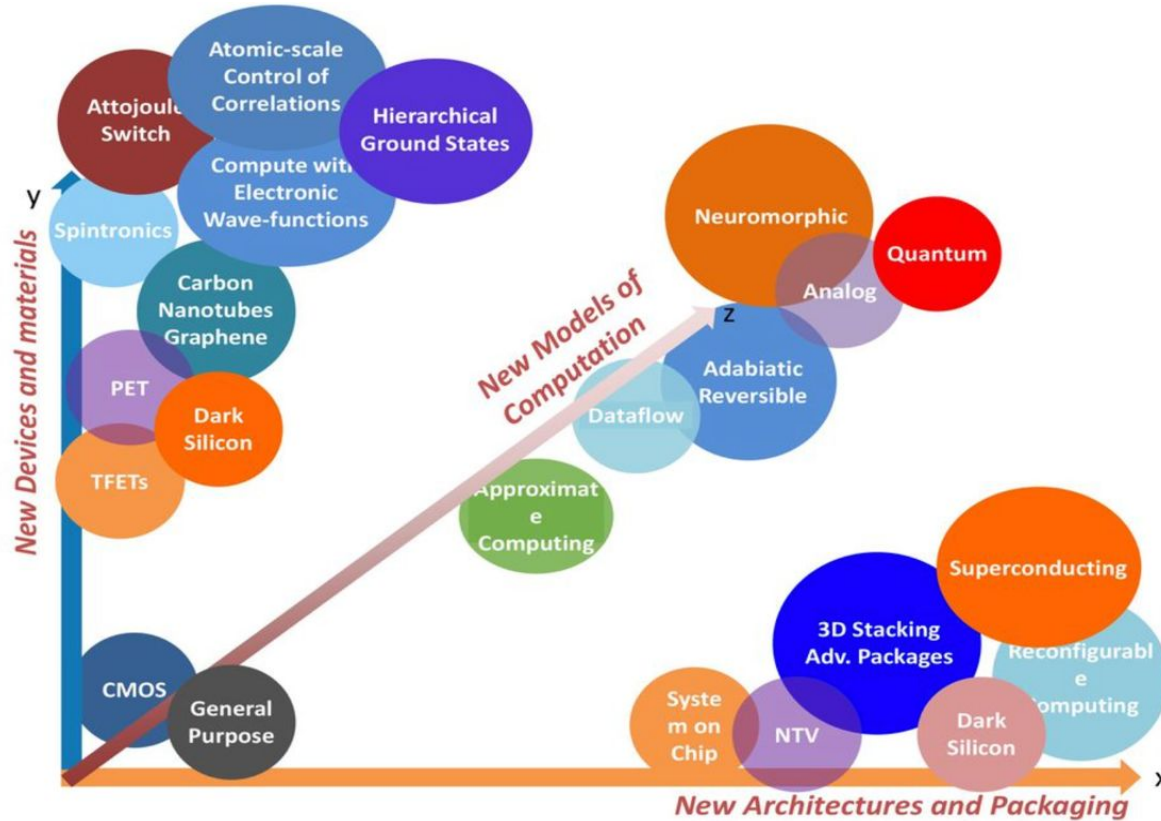
Moore's Law Scaling

(more flattening ahead)



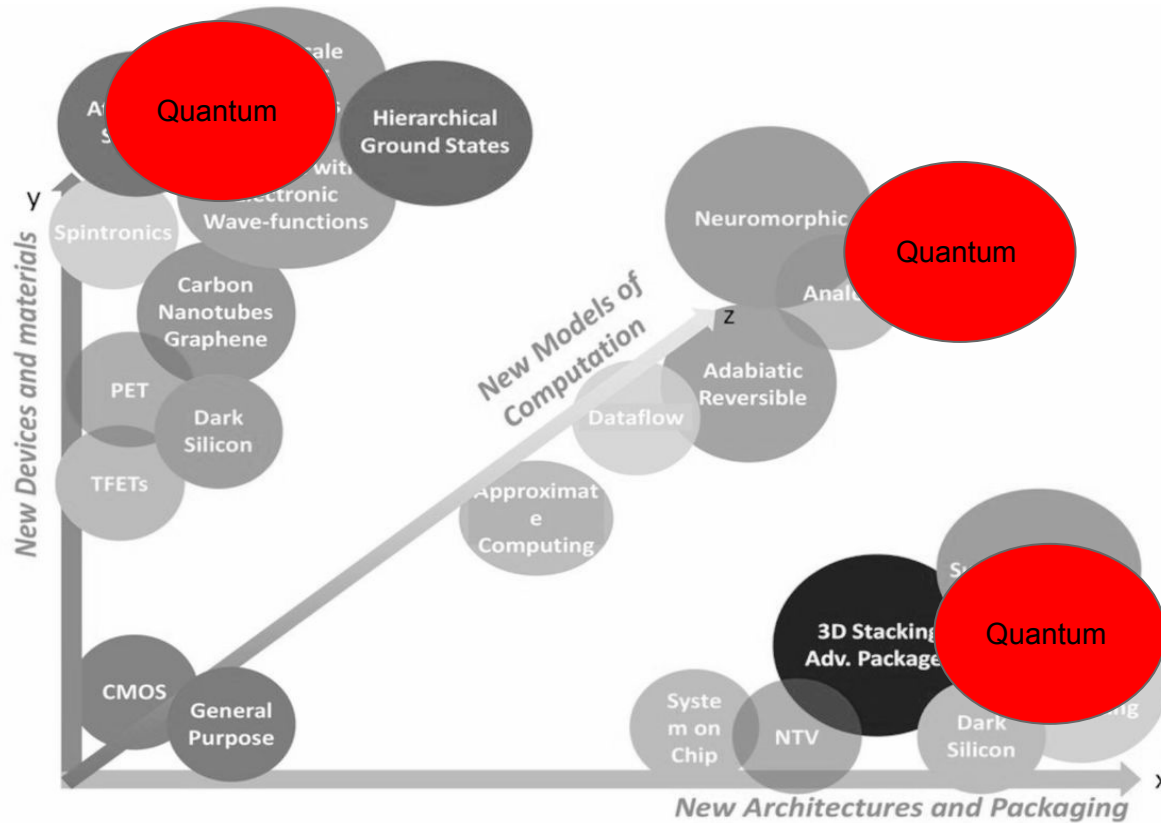
Source: Courtesy of Kunle Olukoton, Lance Hammond, Herb Sutter and Burton Smith

Path Forwards



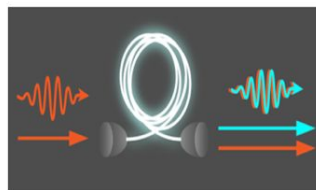
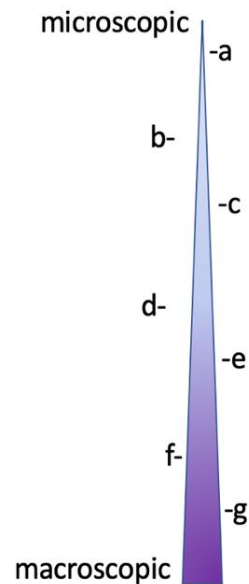
J.Shalf "The future of computing beyond Moore's Law" 2020

Path Forwards

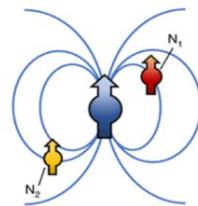


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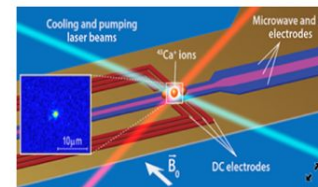
Qubit's Technology Diversity



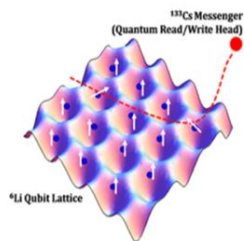
(a) Photon Qubits



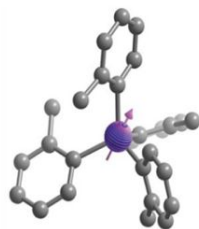
(b) Nuclear Spin



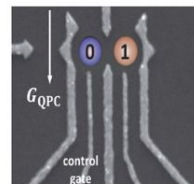
(c) Traps ions



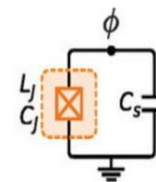
(d) Neutral Atoms



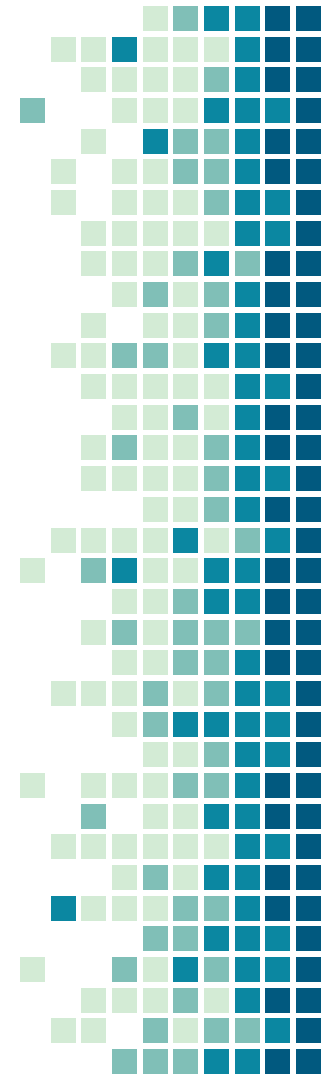
(e) Molecules



(f) Quantum dots

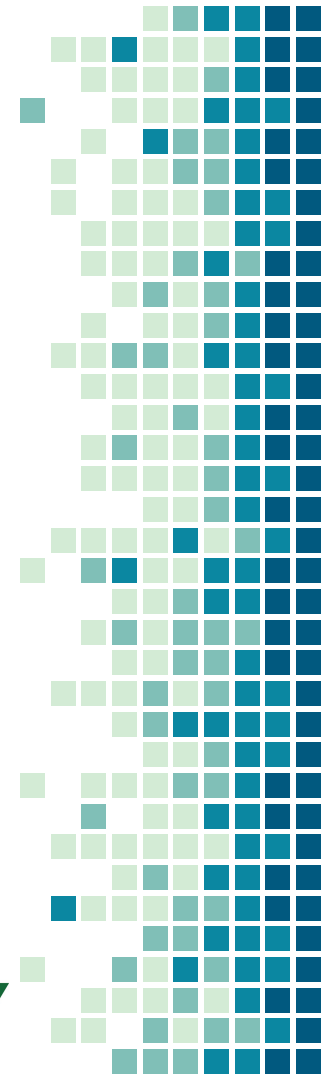


(g) Superconducting circuit



Outline

- Short-term quantum control solutions
 - Custom architectures
 - FPGA-based solutions
- Long-term quantum control solutions
 - New technologies



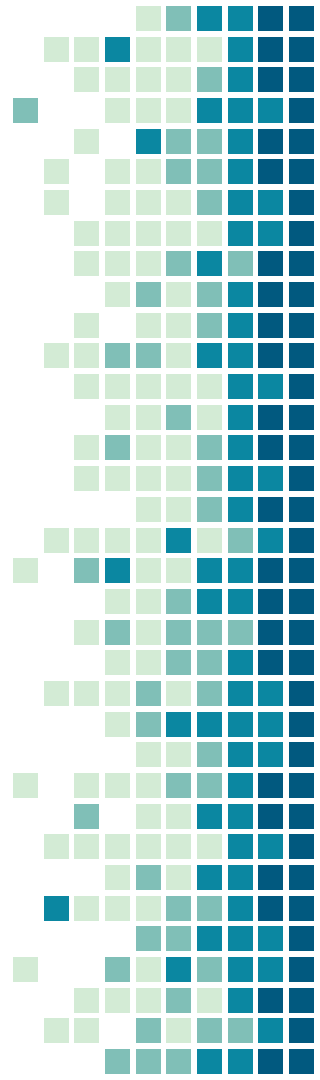
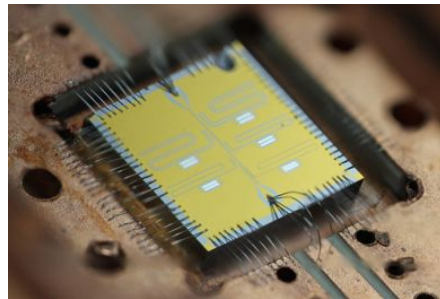
Challenges

Control hardware

Quantum computer vs. accelerator

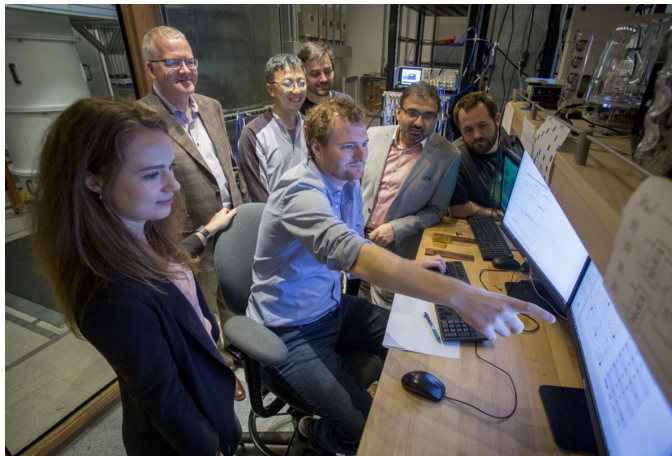
Inaccuracy of terminology

- Quantum computer $\neq$ Quantum Processor



Advanced Quantum Testbed (AQT)

- DOE funded cross-disciplinary project
 - quantum physicists (QNL, MIT LL)
 - material scientists (MF)
 - computer scientists (CRD)
 - engineers (ATAP)
 - industry partners



Controls, controls, and more controls...

Zurich
Instruments

QUASAR/
QubiC

QICK on ZCU-111 &
integrated AFE board



- A mixture of commercial and open-source
- Not shown here:
 - Keysight
 - Qblox
 - Etc
- No 'perfect', comprehensive solution to meet all experimental needs.

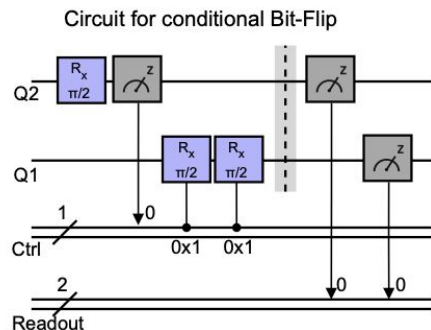
Complexity Nature

Quantum Experiment

Mid-Circuit Measurement

Crucial Operation: *Demonstrating quantum control and calibration*

- **Enabling quantum error correction:** a crucial component of quantum error correction protocols, such as surface codes and Shor codes, which require measuring qubits during computation.
- **Implementing quantum algorithms:** necessary for implementing various quantum algorithms, including quantum teleportation, superdense coding, and quantum simulation.
- **Improving quantum circuit fidelity:** Mid-circuit measurements can help improve quantum circuit fidelity by allowing for the correction of errors that occur during computation.

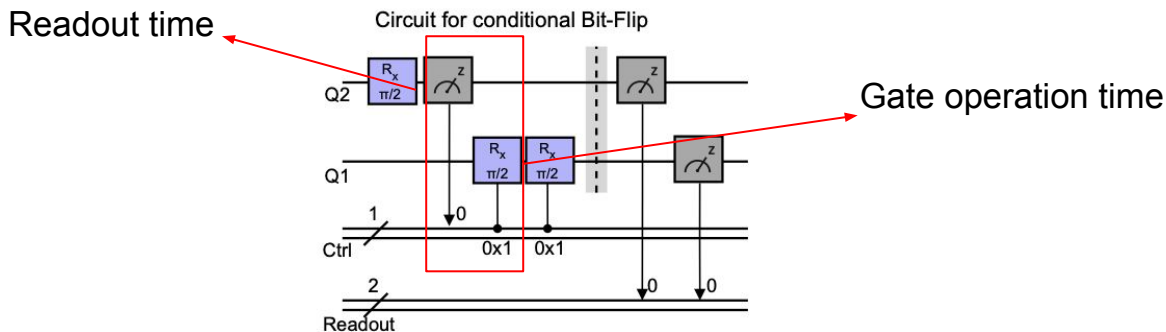


Vora, N. R. et al., "ML-Powered FPGA-based Real-Time Quantum State Discrimination Enabling Mid-circuit Measurements", *arXiv e-prints*, Art. no. arXiv:2406.18807, 2024. doi:10.48550/arXiv.2406.18807.

Mid-Circuit Measurement

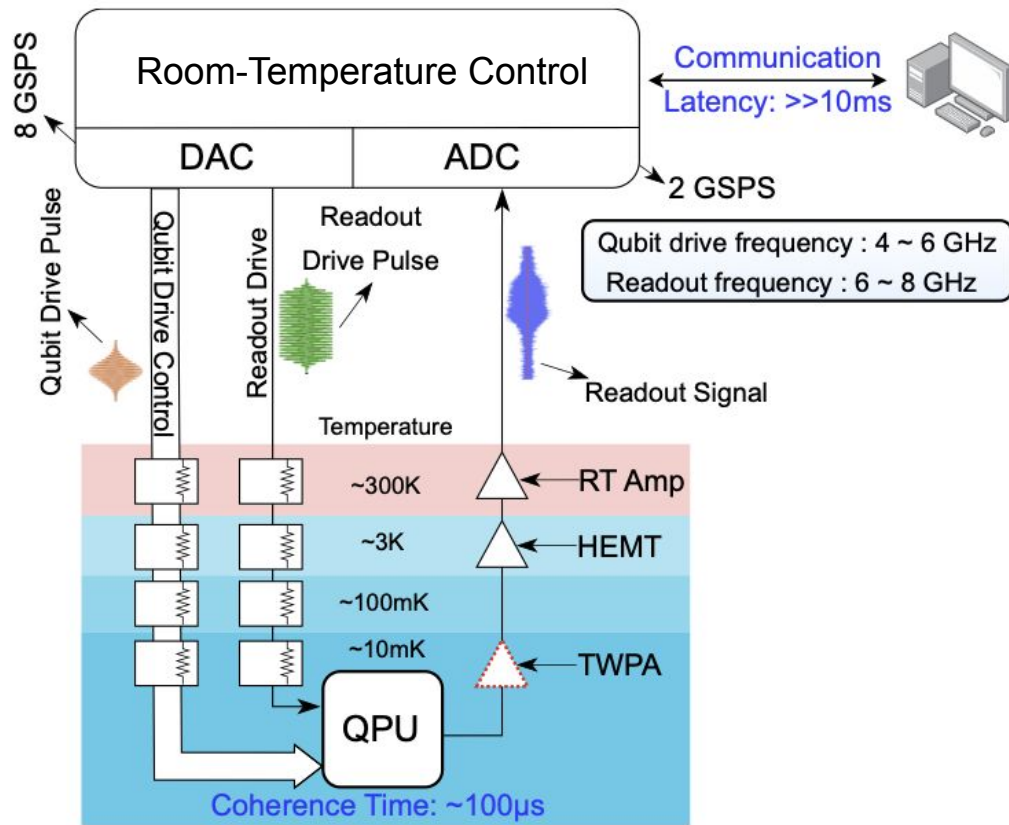
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Control Architecture

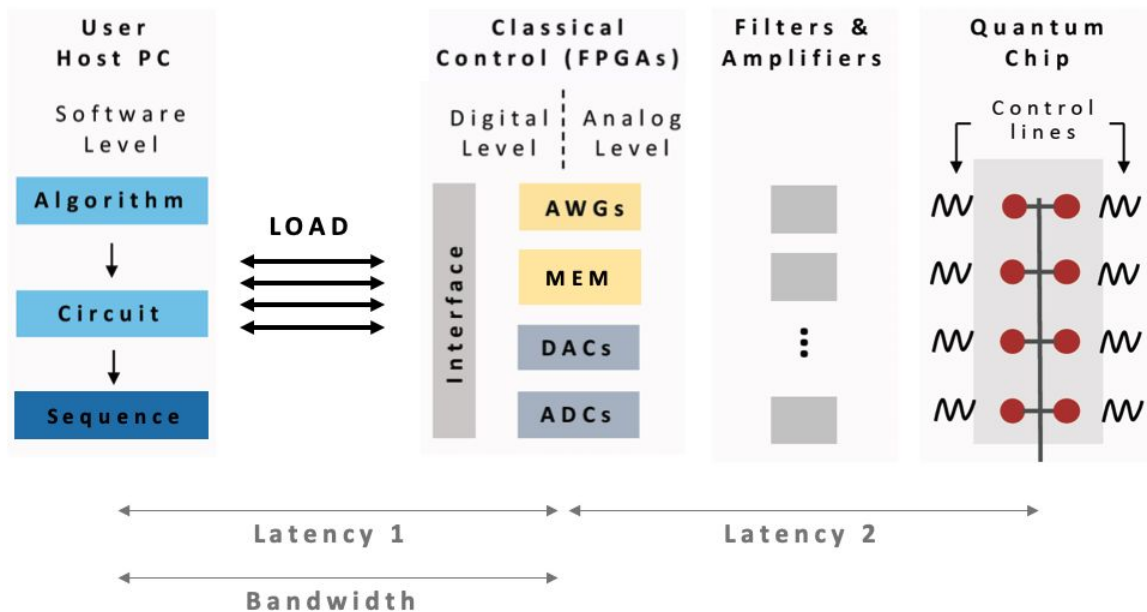


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Control System

Quantum ISA

Software Control



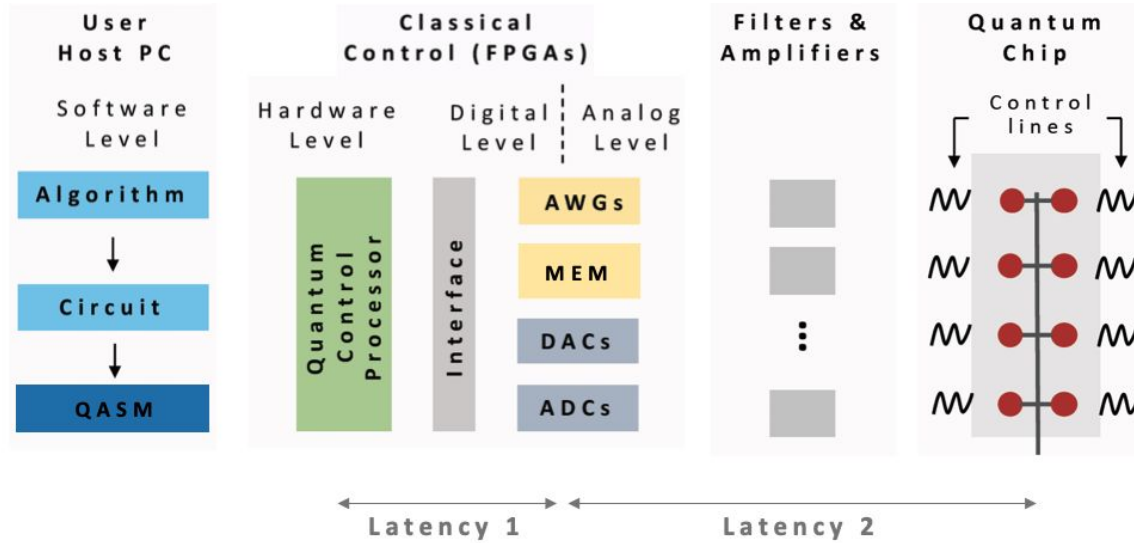
- Everything in one place
- Easier to implement and modify
- NISQ experiments



- Long latency
- Limited bandwidth
- Limited flexibility (bit granularity)
- Poor scalability



Hardware Control



- Reduced latency
- Fast feedback
- Extended functionality in-place
- Potential scalability
- Separation of concepts
- Beyond-NISQ

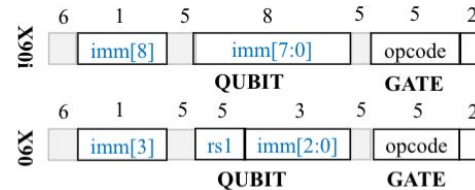


- Requires specialized solution
- Harder to implement and modify
- Requires commercial support

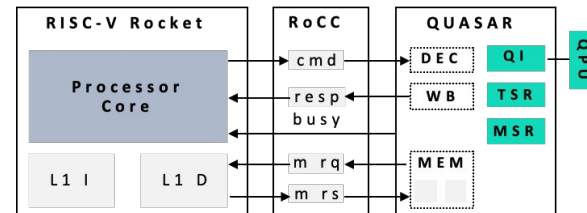


Quantum ISA (QUASAR)

- Quantum Instruction Set Architecture (QUASAR)*
 - extension to RISC-V ISA (open-source, modular, active community, eco-system)
 - supports quantum operations, timing control, etc.
 - transparent, adaptable, open **
- RoCC co-processor adaptation
 - Existing software support
 - Modular approach easily interactable into a big system
 - Low-level customization and flexibility



X90i QUBIT_I;
ADDI x3, 0, QUBIT_MASK;
Y180 x3;
CNOT QUBIT_I_0, QUBIT_I_1;
TSi TIME_STAMP;



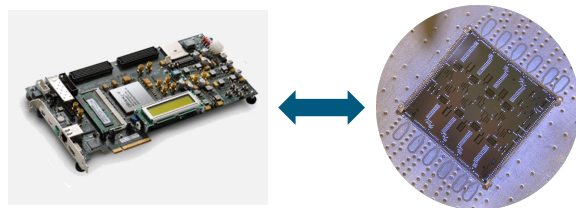
* A. Butko, et al. "Understanding quantum control processor capabilities and limitations through circuit characterization". IEEE ICRC (2020)

** <https://ipo.lbl.gov/quantum-instruction-set-architecture-quasar/>

Experiments with the Quantum Processor

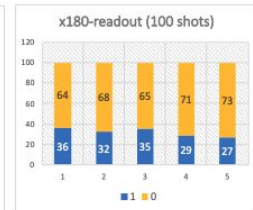
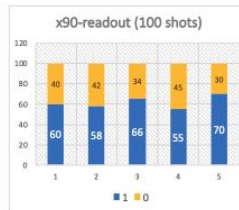
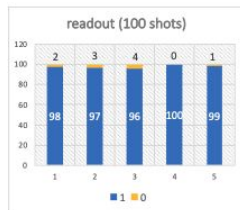
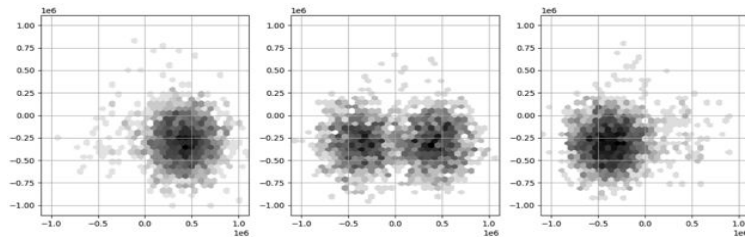
Experimental Setup:

- VC707 FPGA connected to the fridge with the Berkeley QP
- QUASAR-based system deployed on FPGA
- Linux OS running quantum algorithms

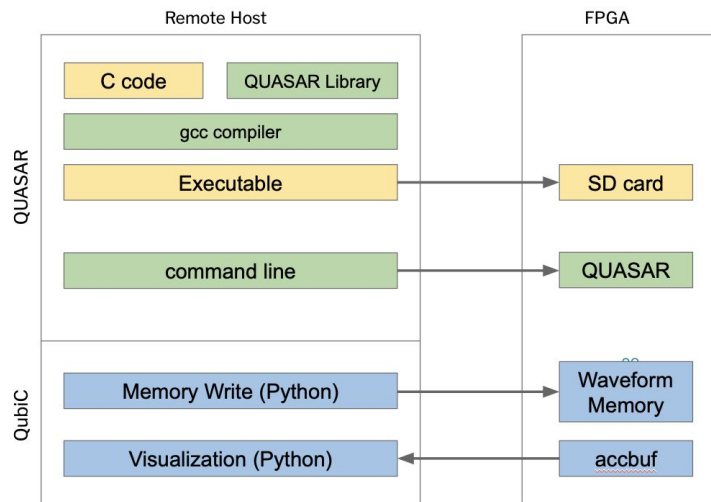


Fast Feedback Demo:

- Mid-circuit measurement
- Statistically significant data collection
- Conditional branching at runtime



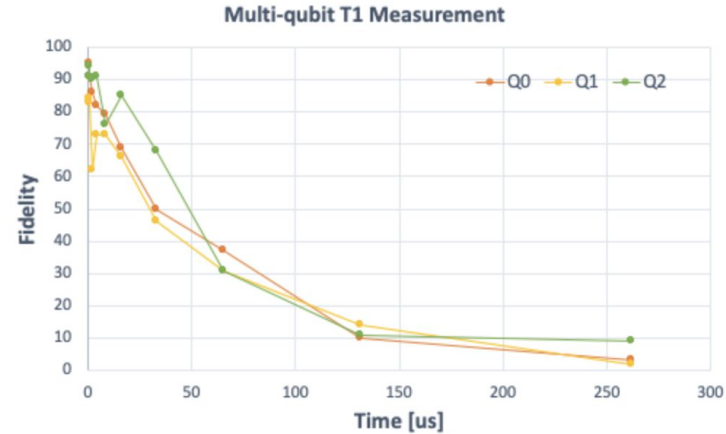
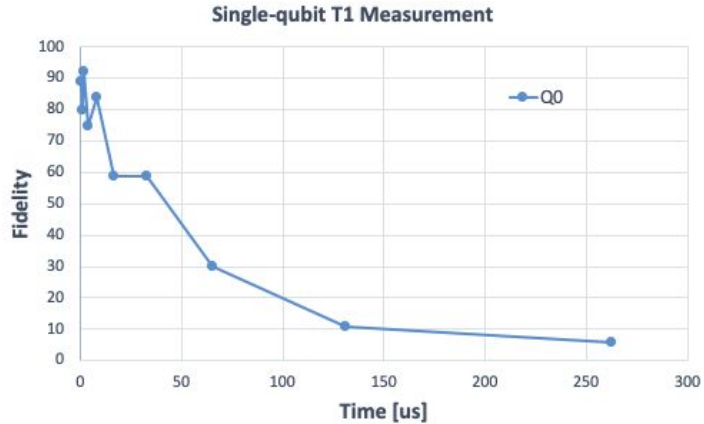
Software Stack



```
#include "include/quasar.h"
int main(int argc, char *argv[]) {
    //Load the entire PLT
    uint64_t data [] = {
        ( 5 << 26 | 32 << 14 | 0 << 2 | 1 ),
        ( 17 << 26 | 40 << 14 | 0 << 2 | 2 );
    };
    for (int i=0; i < 4; i++){ lde(i, data[i]); }
    //Reset HW resources to ZERO
    qReset();
    //Circuit
    y90i(1);
    tsi(8);
    zAi(0, 1234);
    ts(48);
    //Readout pulses
    rdrv(0);
    ts(268);
    read(0);
    //Measurement data
    uint64_t measurement_data;
    mi(measurement_data, 0);
    //Feedback
    if(measurement_data > 0){
        return 0;
    }
}
```

T1 Measurement Results

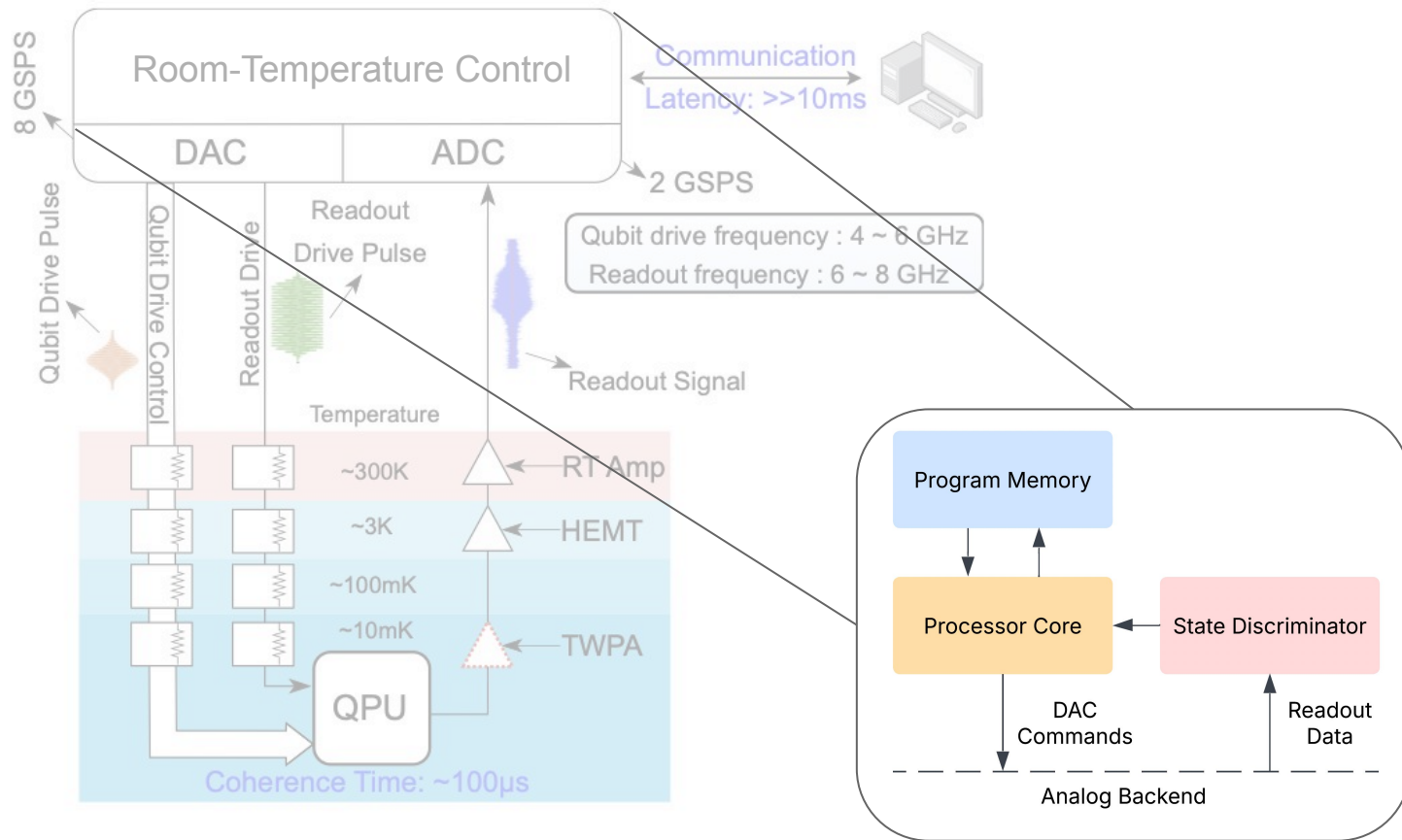
- Sequential vs parallel application
- Single-qubit vs multi-qubit T1 measurement



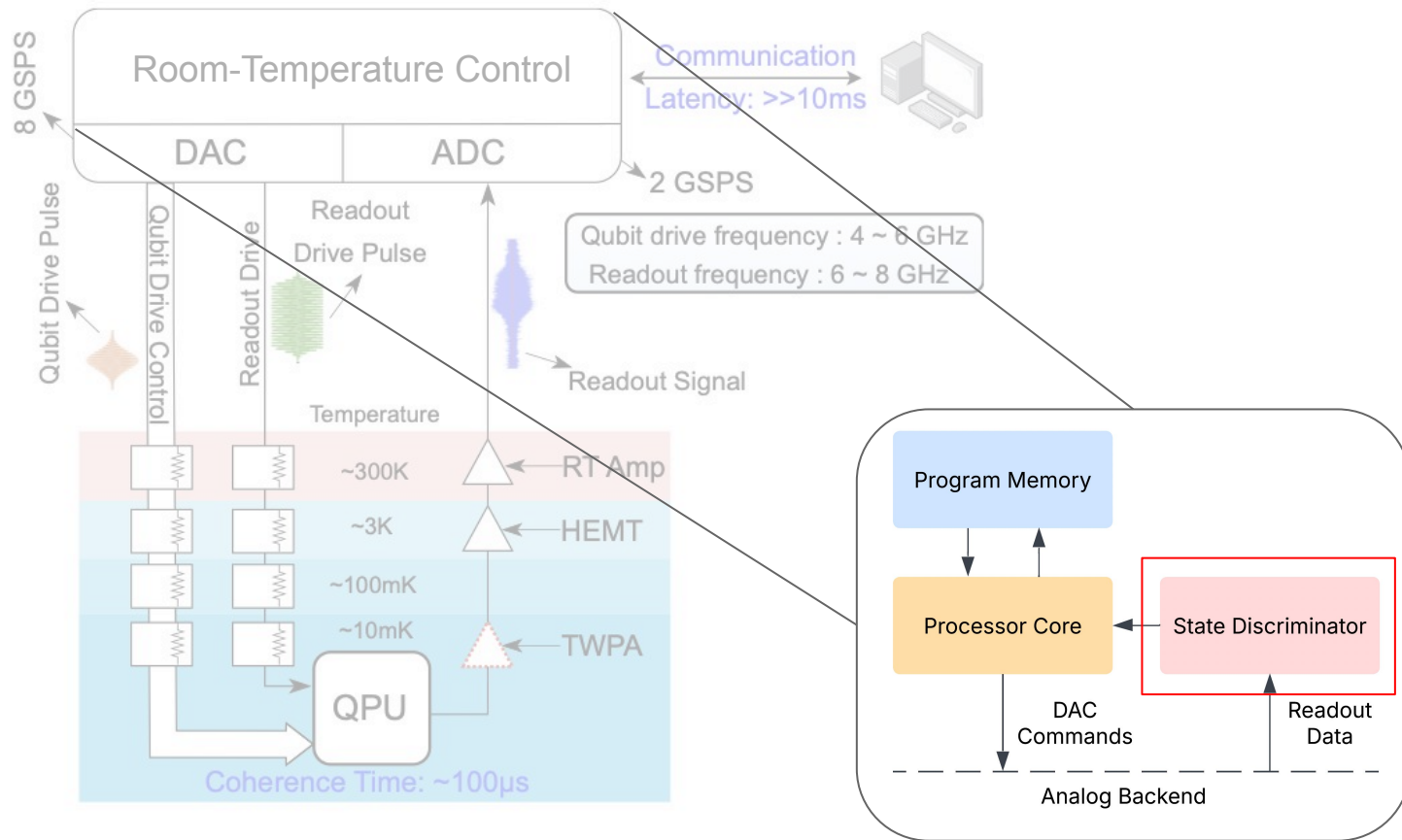
Control System

QSD

Control Architecture



Control Architecture



State of the Art

Quantum State Discriminator

Work	Real-time	Optimization	ML-based	Mid-circuit	Large-scale
Benjamin [1]	No	No	Yes	No	-
Satvik [2]	No	Partially	Yes	No	-
IBM [3]	Yes	No	No	Yes	-
Piero [4]	No	No	Yes	No	-
QubicML [5]	Yes	Yes	Yes	Yes	No

[1] B. Lienhard and e. a. Vepsäläinen. Deep-neural-network discrimination of multiplexed superconducting-qubit states. *Phys. Rev. Appl.*, 17:014024, Jan 2022.

[2] S. Maurya, C. N. Mude, W. D. Oliver, B. Lienhard, and S. Tannu. Scaling qubit readout with hardware efficient machine learning architectures. In *Proceedings of the 50th Annual International Symposium on Computer Architecture, ISCA '23*, New York, NY, USA, 2023. Association for Computing Machinery.

[3] E. Bäumer, V. Tripathi, A. Seif, D. Lidar, and D. S. Wang. Quantum fourier transform using dynamic circuits. 2024.

[4] P. Luchi, P. E. Trevisanutto, A. Roggero, J. L. DuBois, Y. J. Rosen, F. Turro, V. Amitrano, and F. Pederiva. Enhancing qubit readout with autoencoders. *Physical Review Applied*, 20(1), July 2023.

[5] Vora, N. R. et al., “ML-Powered FPGA-based Real-Time Quantum State Discrimination Enabling Mid-circuit Measurements”, *arXiv e-prints*, Art. no. arXiv:2406.18807, 2024. doi:10.48550/arXiv.2406.18807.

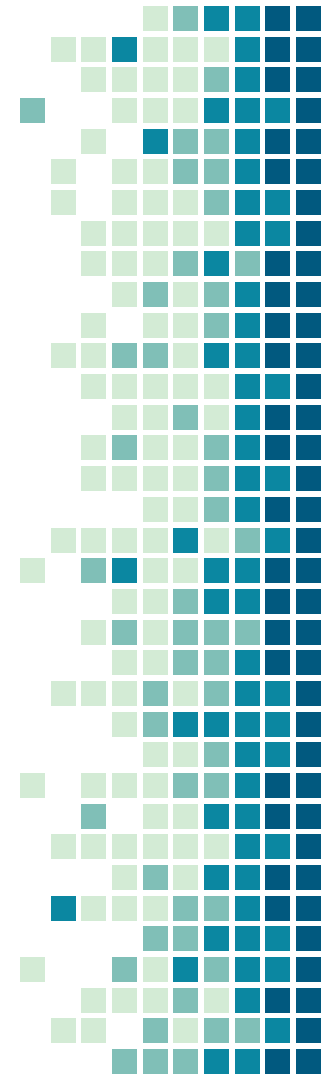
AI capabilities on modern FPGAs

Control Complexity

AMD Xilinx Board Resources

Versal AI Core Series - VCK190

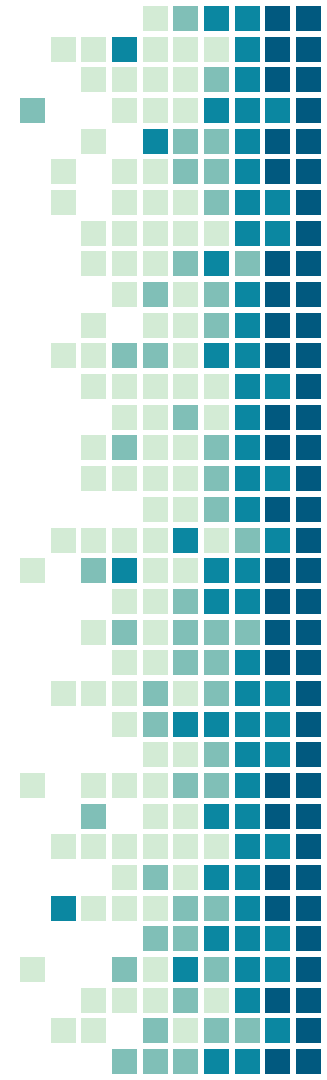
	Programmable Logic (PL)
Customization	Extremely high , allows for custom data paths, processing elements, and control logic.
Performance	Moderate to High , depending on design complexity.
Energy Efficiency	Variable , depends on the specific design; generally higher due to flexibility.
Usability	Complex and requires deep hardware design knowledge; steep learning curve.



AMD Xilinx Board Resources

Versal AI Core Series - VCK190

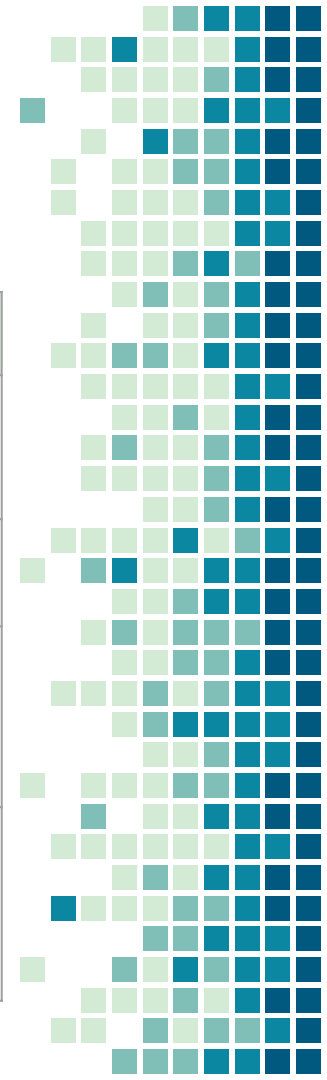
	Programmable Logic (PL)	AI Engine (AIE)
Customization	Extremely high , allows for custom data paths, processing elements, and control logic.	Limited within the scope of AI/ML operations.
Performance	Moderate to High , depending on design complexity.	Very High , optimized for parallel AI operations.
Energy Efficiency	Variable , depends on the specific design; generally higher due to flexibility.	Optimized for energy efficiency in AI workloads; lower than PL for similar tasks.
Usability	Complex and requires deep hardware design knowledge; steep learning curve.	Moderate usability; easier to use with AI frameworks, but still requires understanding of AI models.



AMD Xilinx Board Resources

Versal AI Core Series - VCK190

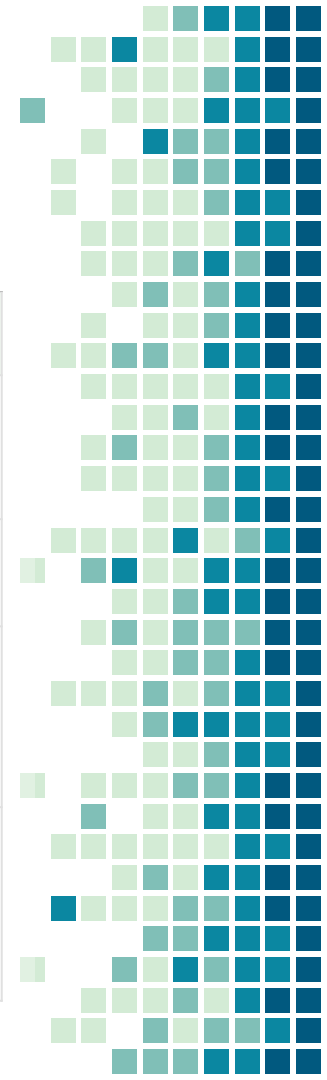
	Programmable Logic (PL)	AI Engine (AIE)	Deep Learning Processing Unit (DPU)
Customization	Extremely high , allows for custom data paths, processing elements, and control logic.	Limited within the scope of AI/ML operations.	Limited to deep learning operations, with focus on neural network acceleration.
Performance	Moderate to High , depending on design complexity.	Very High , optimized for parallel AI operations.	High , specifically optimized for deep learning tasks.
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AMD Xilinx Board Resources

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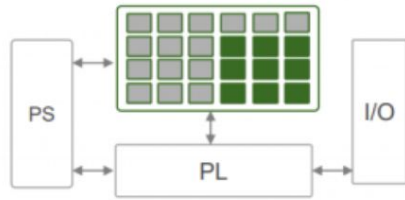
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Architecture

VCK190/AI Engine

AI Engine Array Architecture



Modular and scalable architecture

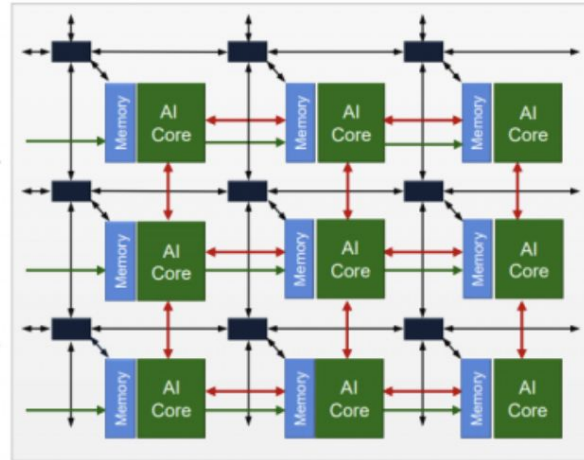
- More tiles = more compute
- Up to 400 per device
 - Versal AI Core VC1902 device

Distributed memory hierarchy

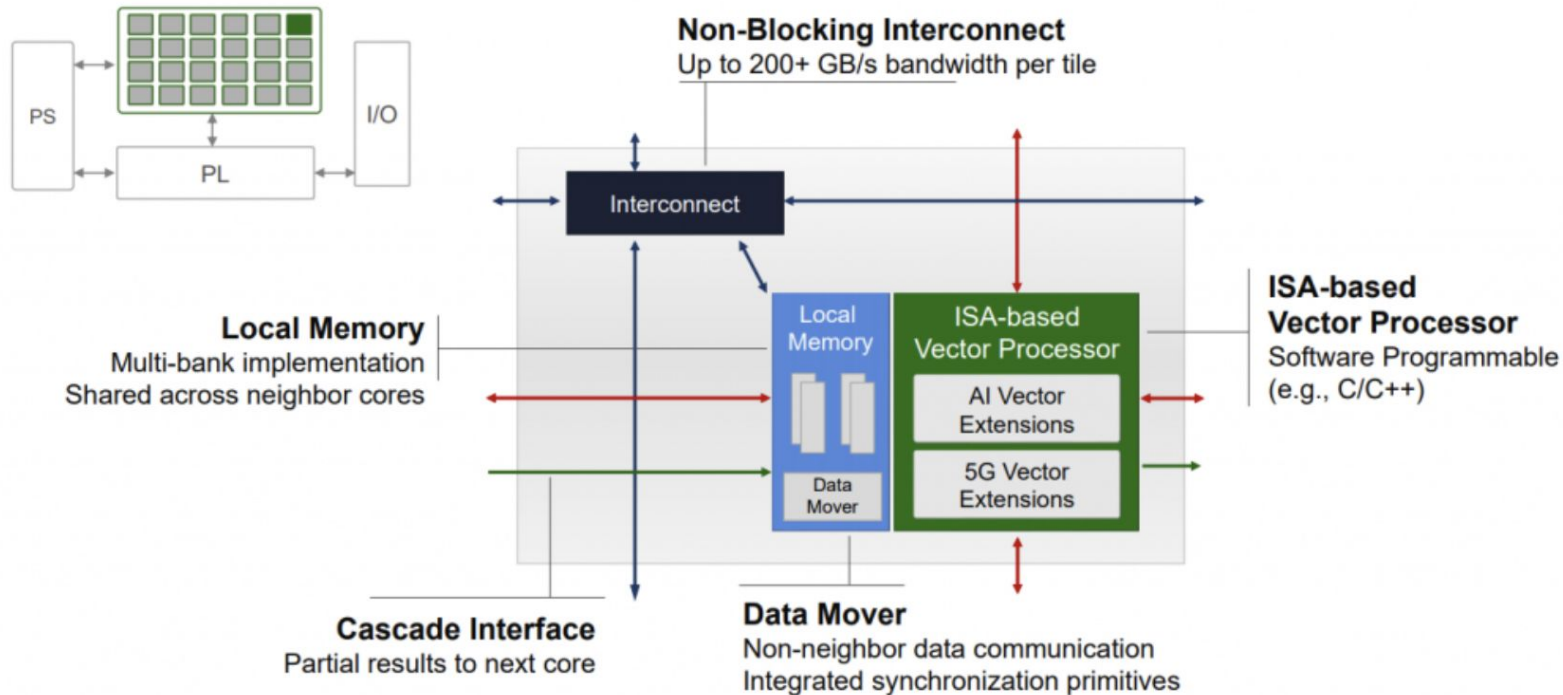
Maximize memory bandwidth

Array of AI Engines

- Increase in compute, memory and communication bandwidth

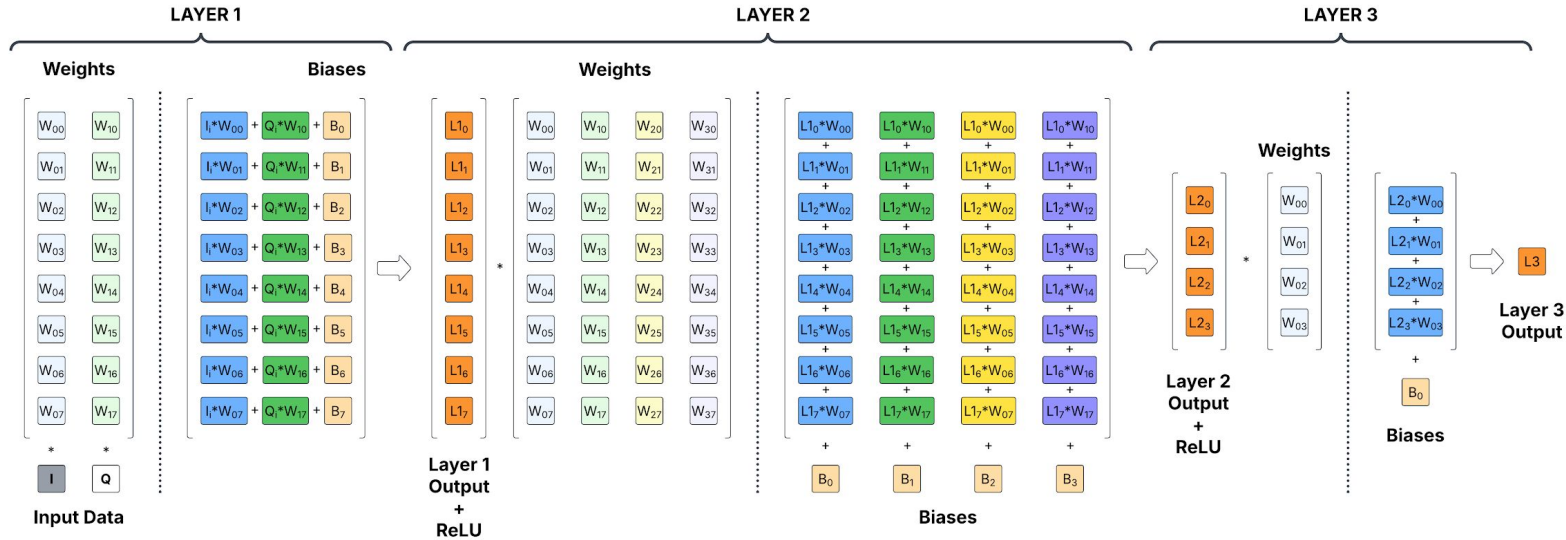


AI Engine Array Architecture



Architecture Implementation

NN with 3 Layers

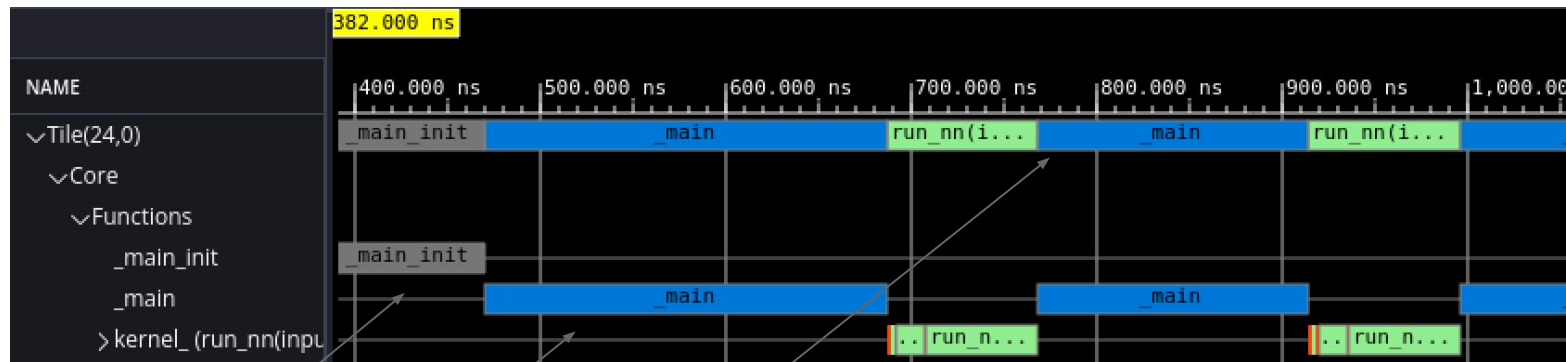


Performance & Power Consumption

Results

Performance Results

AI Engine @ 1250MHz



87.2ns

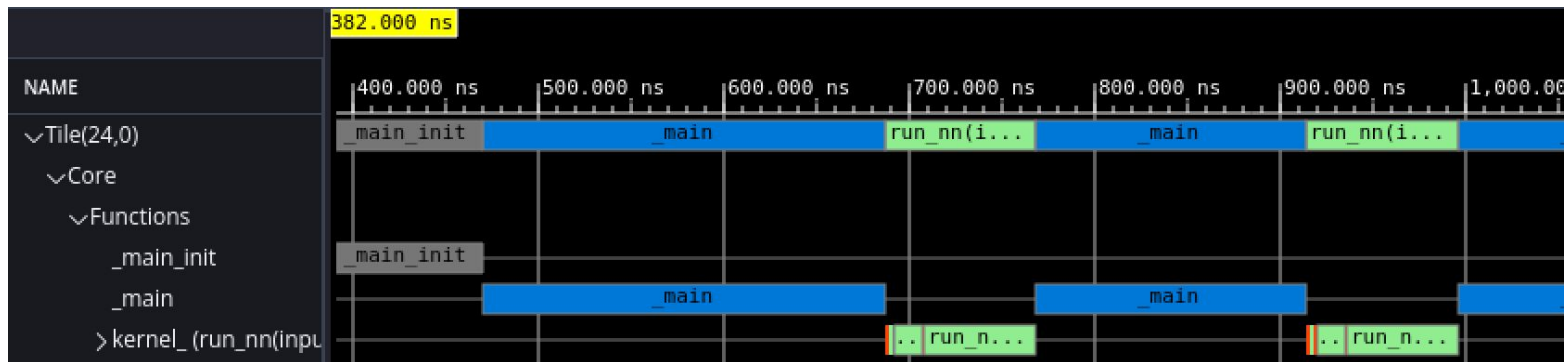
216.8ns

145.6ns*

Kernel: 81.6ns

Performance Results

AI Engine @ 1250MHz

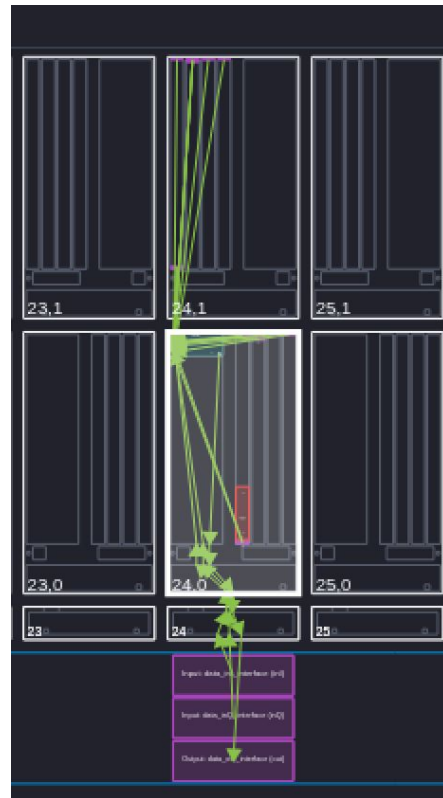


Kernel: 81.6ns

QubiCML:
27 clock cycles,
54ns inference time

Resource Utilization

AI Engine @ 1250MHz



0.25% Kernel Tiles

0.50% Buffer Tiles

0.44% Stream Interconnect Tiles

Power Consumption Estimation

AI Engine @ 1250MHz

Utilization				
Type	Available	Used	Utilization	Power
Cores	400	1	0.25 %	0.593 W
PL	546	0	0.00 %	0.000 W
NoC	32	0	0.00 %	0.000 W

Power		
Supply	Voltage	Power
VCCINT	0.800 V	0.564 W
VCCAUX	1.500 V	0.029 W
Total		0.593 W

Dynamic Power	
Type	Value
AIE Core	0.092 W
Memory	0.501 W

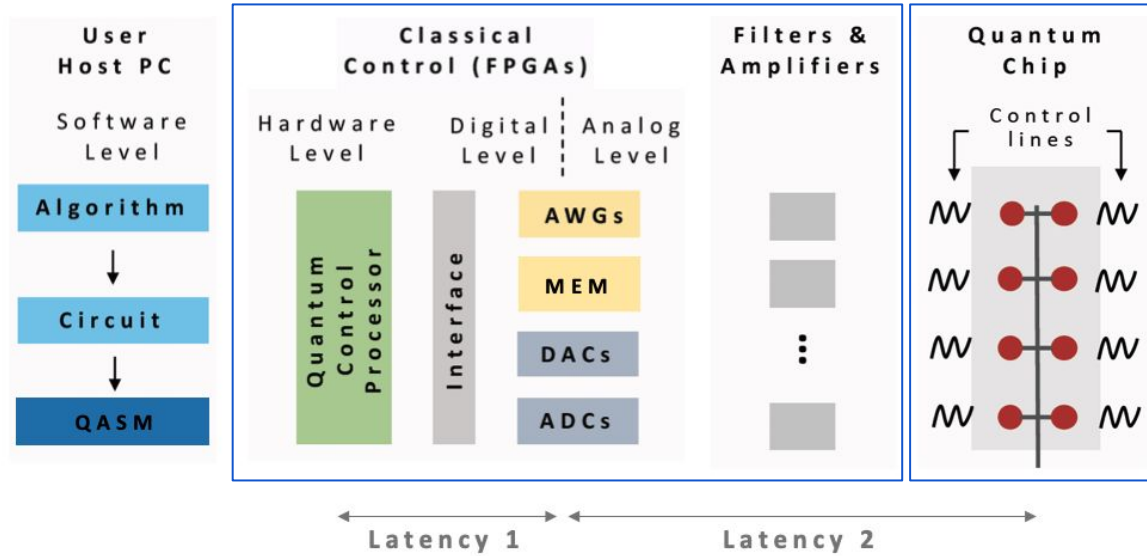
Power		
Total	5.246 W	
XCVC1902-VIVA1596-2MP-E-S	5.246 W	99.99 %
Active	0.840 W	16.02 %
Static	4.405 W	83.98 %

Future Systems

Low-Temperature Control

What about this?

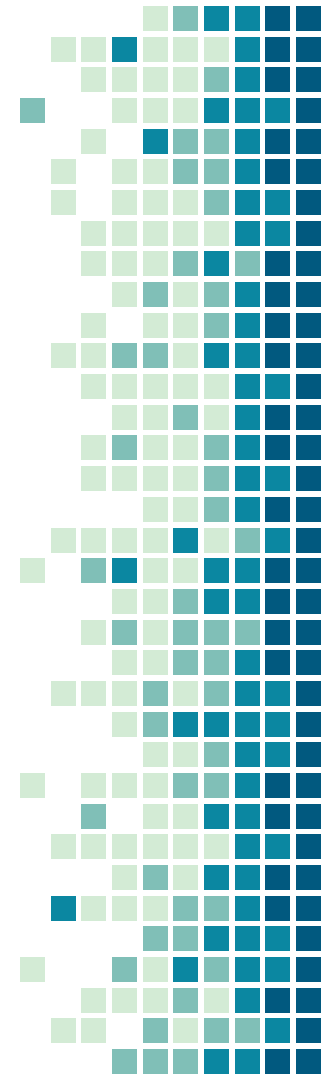
Low temperature



- Reduced latency further
- Super Fast feedback
- Integrated scalability
- Far Beyond-NISQ



- Potential noise
- Requires new technologies
- New architectures



New Trends: Low-temperature Technologies

B. Patra et al., "A Scalable Cryo-CMOS 2-to-20GHz Digitally Intensive Controller for 4×32 Frequency Multiplexed Spin Qubits/Transmons in 22nm FinFET Technology for Quantum Computers," 2020 IEEE International Solid-State Circuits Conference - (ISSCC), 2020

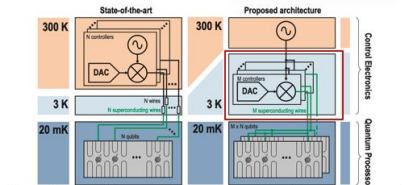
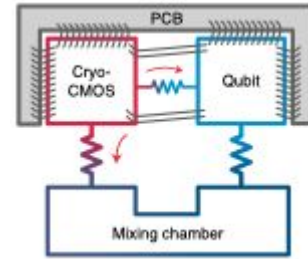
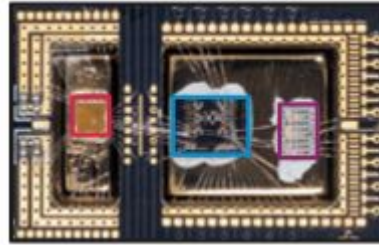
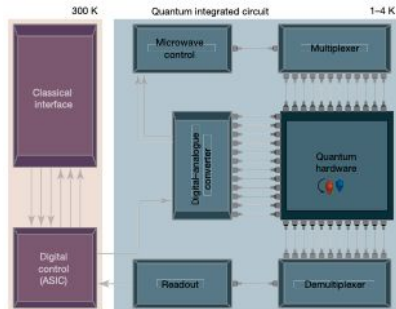


Figure 19.1.1: Qubit control signals, current state-of-the-art controller and presented cryogenic controller with frequency multiplexing.

S. J. Pauka et al., "A cryogenic cmos chip for generating control signals for multiple qubits," Nature Electronics, 2021.



L. Petit et al., "Universal quantum logic in hot silicon qubits," Nature, 2020.

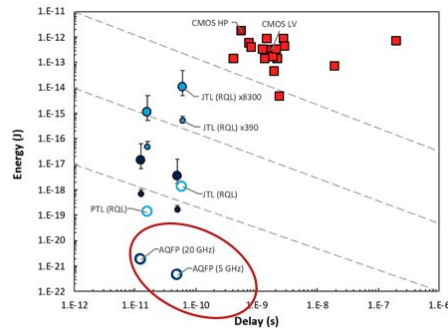


Noise Budget
Power Budget
Monolithic Integration

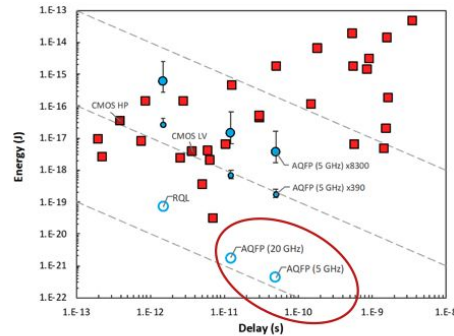
Adiabatic Quantum-Flux-Parametron (AQFP)

- Energy-Efficient Circuits *
- Dynamic energy dissipation is reduced due to the adiabatic switching operations using AC excitation currents
- AQFP could overcome the power/energy dissipation limitation in conventional superconductor logic families such as rapid-single-flux-quantum (RSFQ)

* Chen, O. et al. “Adiabatic Quantum-Flux-Parametron: Towards Building Extremely Energy-Efficient Circuits and Systems.” Scientific Reports 9 (2019): n. pag.



CEQIP-4 Energy versus Delay for Interconnects of 1 mm Length



CEQIP-3 Energy versus Delay for Intrinsic Elements

Circuit Complexity

** Cryogenic Electronics and Quantum Information Processing,” IEEE International Roadmap for Devices and Systems, 2020IRDS_CEQIP, 2020. <https://irds.ieee.org/editions/2020>

Superconducting Logic Research

Opportunity: *Low power, ultra-high performance computing*

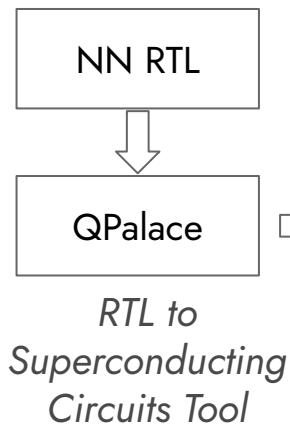
- 600+GHz clock frequencies demonstrated in lab for superconducting RSFQ logic
- Lossless electrical data transmission
- Lower power (even with cryo-refrigeration taken into account)

Active Research Directions

- **SuperTools:** Develop superconducting variant of RISC-V processor to evaluate emerging Superconducting Electronic Design Tools for iARPA.



Map the design to Superconducting Computing

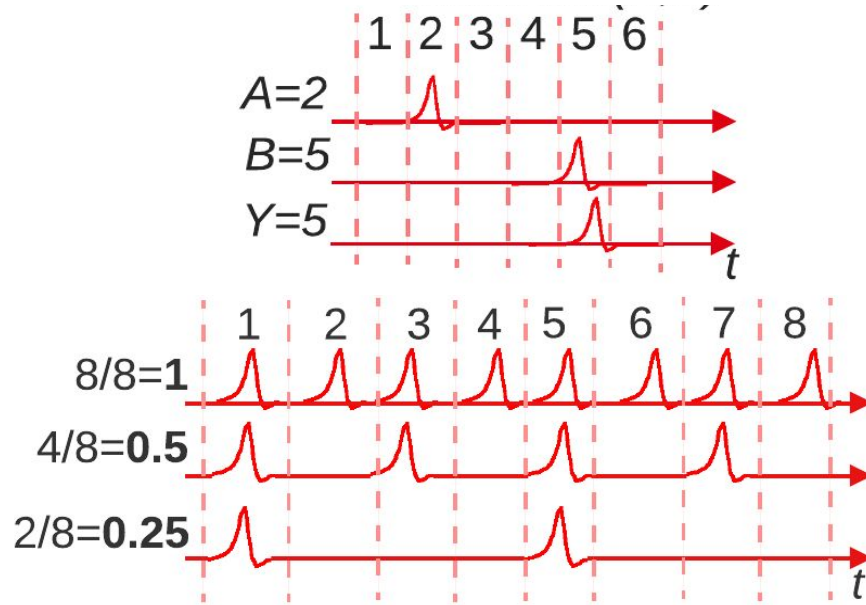


Design	Frequency of Operation	Latency	Hardware	Power Consumption	Number of JJs
ML In FPGA	500MHz	54 ns	FPGA	-	-
ML in eRSFQ	30 GHZ	0.9 ns	SuperC. Chip	5 mW	344K JJs



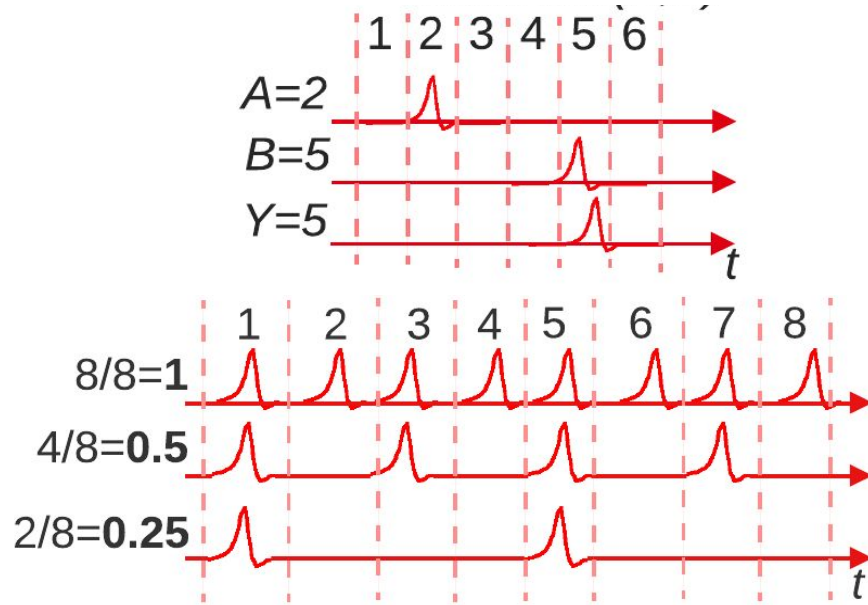
- 98% Faster than state-of-the-art
- 0.5% of the 1W power budget
- Largest Demonstrated Superconducting Chip is a single multiplier with ~20K JJs

Use Biologically Inspired Data Representation to Map Information : Unary-SFQ (U-SFQ)

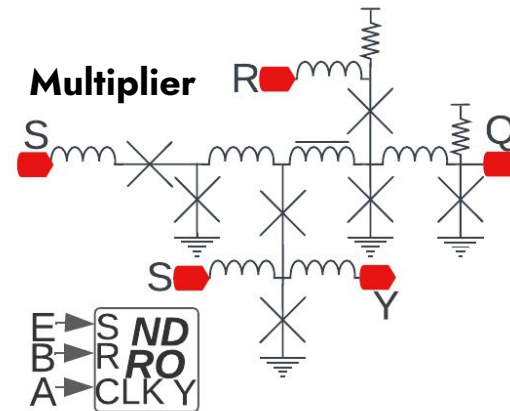
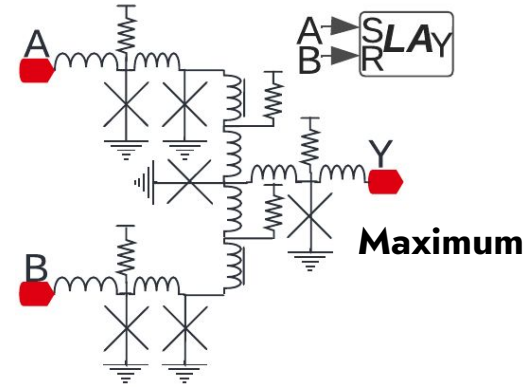


Gonzalez-Guerrero, *et.al.*, Temporal and SFQ Pulse Streams
Encoding for Area-Efficient Superconducting Accelerators.
ASPLOS'22

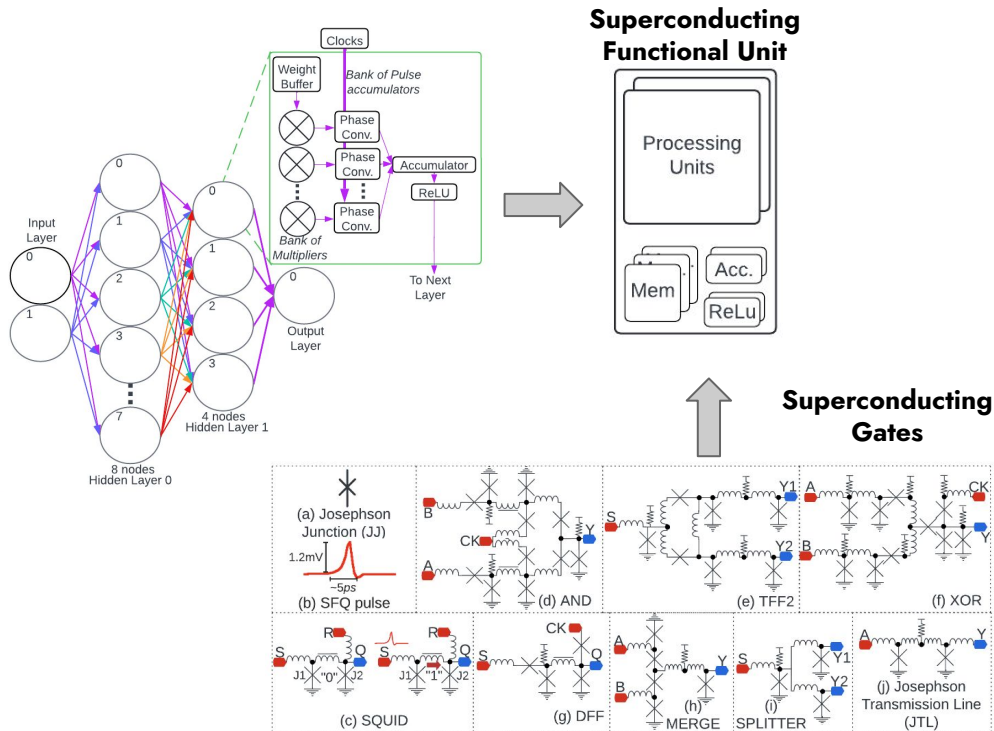
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Practical Superconductor Circuit for Qubit State Discrimination



- **Superconducting Quantum State Discriminator:** functional units connected through a crossbar
- **Functional Unit:** multipliers, accumulators, memory banks, and ReLU units.
- **Cell Library:** neurons, synapses, memory banks

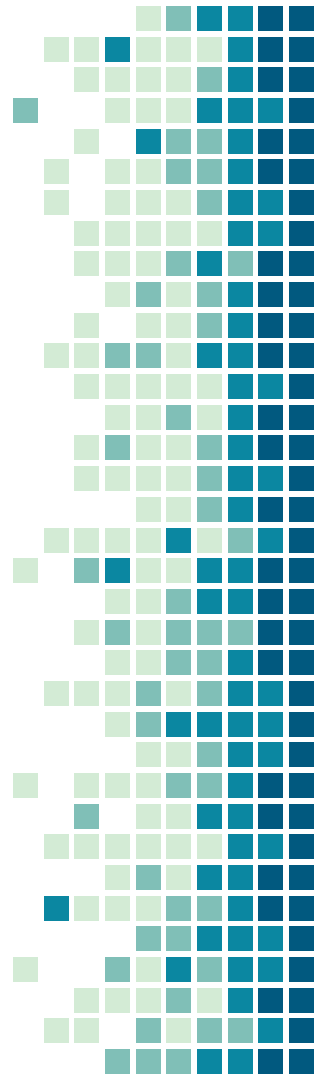
Practical Superconductor ML accelerator for Qubit state discrimination

Design	Number of Bits	Frequency of Operation	Information Encoding	Latency	Hardware	Power Consumption	Number of JJs
ML in FPGA	28	500MHz	Binary	54 ns	FPGA	-	-
ML in eRSFQ	8	30 GHZ	Binary	0.9 ns	SuperC	5 mW	344K JJs
ML in eRSFQ	4	50 GHz	Unary	4.32 ns	SuperC	0.09mW	3K JJs

- 92% Faster than state-of-the-art
- 98% less power than Conventional SFQ computing
 - 99% less JJs (area)
 - Under 20K JJs -> Practical!

Nature paper coming shortly

"Cryogenic hardware accelerator for
Quantum State Discrimination at 4K",
P. Gonzalez-Guerrero et al.



Thank you for your attention.
Questions?

